

Mert Toslali

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Staff Research Scientist at IBM Research, working on AI, cloud systems, and software performance. Received his Ph.D. in Computer Engineering from Boston University. His research focuses on using AI and automated experimentation to make complex computing systems more efficient and reliable.

Education

Boston University

Ph.D., Computer Engineering, GPA: 4.0/4.0

Boston, MA, USA

Sept 2018 - May 2023

- Thesis: Efficient navigation of performance unpredictability in cloud through automated analytics systems
- Award: Distinguished Computer Engineering Fellowship

Bilkent University

B.S., Electrical and Electronics Engineering, GPA: 3.63/4.0

Turkey

Sept 2011 - June 2016

- Award: Comprehensive Scholarship (100%)

University of Nebraska Lincoln

Exchange Student, Electrical and Computer Engineering

Lincoln, NE, USA

Aug 2014 - Jan 2015

Professional experience

IBM T.J. Watson Research Center

Staff Research Scientist (Manager: Dr. Srinivasan Parthasarathy)

Cambridge, MA, USA

June 2023 - present

- **Technical lead** for the IBM Research challenge on LLM training and inference optimization, developing an AI-native discovery loop to automatically discover and validate system policies before real-world deployment.
 - **Inference optimization:** Contributed policies discovered through the AI-native loop to the CNCF llm-d project, reducing critical-traffic tail latency by up to 98%.
 - **Training optimization:** Contributed a co-located training architecture integrated into Hugging Face TRL, achieving 2–3× faster post-RL training and adopted as the default.
- **Technical lead** for the IBM Research challenge on software and vendor risk analysis, leveraging generative AI to accelerate risk assessments and strengthen national cybersecurity.

IBM T.J. Watson Research Center

Research Intern, 4x summer internships (Manager: Dr. Srinivasan Parthasarathy)

Yorktown Heights, NY, USA

May 2018 - Jul 2022

- Developed automated, machine learning-driven methods aimed at significantly enhancing efficiency in diagnosing application performance and deploying new code within cloud environments.

PeacLab, Boston University

Research Assistant (Advisor: Prof. Ayse K. Coskun)

Boston, MA, USA

Sept 2018 - May 2023

- Developed automated analytics approaches using statistically driven techniques to accurately localize and diagnose performance variations in code and help prevent them in code delivery cycles.

CyanGate

Senior Software Engineer

Ankara, Turkey

Jan 2018 - July 2018

- Developed file distribution platform integration with Google Drive, Box, Amazon S3

OpsGenie-Atlassian

Software Engineer

Ankara, Turkey

July 2017 - Dec 2017

- Developed scalable backend services, monitoring, and logging systems using Java, ANTLR, Elasticsearch, and AWS (Lambda, Kinesis, DynamoDB, SQS, EC2, ElastiCache, Redis).

Turkcell

Software Engineer

Istanbul, Turkey

July 2016 - July 2017

- Developed REST API services for the BiP messaging app (Translate, Pharmacy, Weather, Face Recognition), the OTA client for SIM card applications, and BiP's payment reconciliation infrastructure.

Seneka

Software Engineer, Part time

Ankara, Turkey

Sept 2015 - July 2016

- Contributed to development of Electronic Document Management System for Turkish Ministry

Turkcell

Software Engineering Intern

Ankara, Turkey

July 2015 - Sept 2016

- Contributed to a research project for energy-aware data center engineering

Publications

- D. Guhathakurta*, **M. Toslali***[†], S. Parthasarathy, J. Chen, M. Kalantar, N. Masluk, V. Ramani, A. Tantawi, F. Oliveira, “Simulate Before You Scale: High-Fidelity LLM Inference Planning Without GPUs”. In Proc. ACM Meas. Anal. Comput. Syst. (SIGMETRICS’27), Vol. 10, No. 3. (*co-first, [†]corresponding author)
- **M. Toslali**, E. Snible, J. Chen, A. Cha, S. Singh, M. Kalantar, S. Parthasarathy, “AgraBOT: Accelerating Third-Party Security Risk Management in Enterprise Setting through Generative AI”. In ACM International Conference on the Foundations of Software Engineering (FSE’24).
- **[Best paper award] M. Toslali**, S. Qasim, S. Parthasarathy, F. Oliveira, H. Huang, G. Stringhini, Z. Liu, and A. Coskun. “An Online Probabilistic Distributed Tracing System”. In IEEE International Conference on Cloud Engineering (IC2E’24).
- S. Qasim, **M. Toslali**, Q. Clark, S. Parthasarathy, F. Oliveira, A. Liu, G. Stringhini, A. Coskun, “Efficient Navigation of Cloud Performance with ‘nuffTrace”. In IEEE International Conference on Cloud Engineering (IC2E’23)
- **M. Toslali**, E. Ates, D. Huye, Z. Zhang, L. Liu, S. Puterman, A.K. Coskun, R.R. Sambasivan, “VAIF: Variance-driven Automated Instrumentation Framework”. In SIGOPS Oper. Syst. Rev. June 2022.
- **M. Toslali**, S. Parthasarathy, F. Oliveira, H. Huang, A. Coskun, “Iter8: Online Experimentation in the Cloud”. In Proceedings of the ACM Symposium on Cloud Computing (SoCC’21).
- **M. Toslali**, E. Ates, A. Ellis, Z. Zhang, D. Huye, L. Liu, S. Puterman, A.K. Coskun, R.R. Sambasivan, “Automating instrumentation choices for performance problems in distributed applications with VAIF”. In Proceedings of the ACM Symposium on Cloud Computing (SoCC’21).
- S. Allen, **M. Toslali**, F. Oliveira, S. Parthasarathy, A. Coskun, “Tritium: A Cross-layer Analytics System for Enhancing Microservice Rollouts in the Cloud”. In Proceedings of the 7th International Workshop on Container Technologies and Container Clouds (WOC’21).
- **M. Toslali**, S. Parthasarathy, F. Oliveira, A. Coskun, “JACKPOT: Online experimentation of Cloud Microservices”. In 12th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud’20).
- E. Ates, L. Sturmman, **M. Toslali**, O. Krieger, R. Megginson, A. K. Coskun, R. R. Sambasivan, “An automated, cross-layer instrumentation framework for diagnosing performance problems in distributed applications”. In Proceedings of ACM Symposium on Cloud Computing (SoCC’19).

Professional affiliations

- **AAAI Conference on AI, Ethics, and Society (AIES)**, *PC Member*, Reviews: 7 (2024).
- **IEEE International Conference on Big Data (BigData)**, *PC Member*, Reviews: 5 (2024).
- **International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)**, *PC Member*, Reviews: 4 (2024).
- **International Conference on Cloud Computing, GRIDs, and Virtualization (CLOUD)**, *PC Member* (2024).
- **European chapter of ACM SIGOPS (EuroSys)**, *Shadow PC Member*, Reviews: 8 (2022).

Teaching affiliations

- **Boston University**, *Teaching Assistant* in Advanced Data Structure and Algorithms (2019 – 2021).
- **Bilkent University**, *Teaching Assistant* in Computer Science 101, Physics 101, and Physics 102 (2012 – 2014):
- **Bilkent University**, *Volunteer Teacher* in Elementary School Mathematics and Science (2012 – 2014).

Skills

Applied Machine Learning, Large Language Models, LLM Inference & Serving, LLM Training & Post-training, GPU/Accelerator Optimization, Discrete-event Simulation, Agentic Systems, Distributed Tracing, A/B testing, Online experimentation, Cloud Microservices, Software Risk and Compliance, Python, Java, C#, .Net, Kubernetes, Istio, Docker, Amazon Web Services, Huggingface, Elastic Search, SQL, JavaScript, Hibernate, Mongo DB, Oracle DB, Spring, Spring boot.

Open-source software

- **BLIS**: Full-stack LLM inference simulator (CPU-only, high-fidelity, $\sim 200\times$ faster than real execution). github.com/inference-sim/inference-sim
- **NOUS**: Agentic harness for AI-assisted discovery of LLM serving policies; produced the admission- and flow-control policies merged into llm-d. github.com/AI-native-Systems-Research/agentik-strategy-evolution
- **ARCHON**: Architecture-aware validation of AI-generated software changes on real systems. github.com/AI-native-Systems-Research/archon
- **Hugging Face TRL**: Designed a co-located architecture delivering $2\text{--}3\times$ faster post-RL training by sharing GPUs across generation and training; merged and made the default workflow ($\sim 4\text{M}$ monthly, $54\text{M}+$ all-time downloads). github.com/huggingface/trl
- **Iter8**: Open-sourced online experimentation framework; integrated by Kiali, Open Data Hub, and Seldon.

Conference presentations and invited talks

- Conference presentation, AI-assisted discovery for LLM serving (BLIS/NOUS), at KubeCon + CloudNative-Con North America 2026
- Invited talk, GPU-free simulation of Kubernetes-native LLM inference (BLIS), at CNCF-hosted Cloud Native AI + Inference Day 2026
- Conference presentation, Optimizing LLM training in Hugging Face TRL, at PyTorch Conference 2025
- Conference presentation, AgraBOT: Accelerating Third-Party Security Risk Management in Enterprise Setting, at ACM International Conference on the Foundations of Software Engineering (FSE'24)
- Poster presentation, Diagnosing behavioral causes of contention in distributed systems, at USENIX Symposium on Networked Systems Design and Implementation (NSDI '24)
- Invited talk, AI for Cloud operations, at Red Hat Operate First Community Meetup 2023
- Poster presentation, Efficient Navigation of Cloud Performance with 'nuffTrace, at IEEE International Conference on Cloud Engineering (IC2E'23)
- Conference presentation, Online Experimentation for Cloud Applications, at the Conference on Machine Learning and Systems, MLSys'22
- Conference presentation, Automating instrumentation choices for performance problems in distributed applications with VAIF, at ACM Symposium on Cloud Computing, SoCC'21
- Conference presentation, Iter8: Online Experimentation in the Cloud, at ACM Symposium on Cloud Computing, SoCC'21
- Conference presentation, JACKPOT: Online Experimentation of Cloud Microservices, in 12th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud 20)
- Invited talk, Online experimentation in the Cloud, at BU, IBM, RedHat Cloud Computing Workshop, 2020
- Conference presentation, An automated, cross-layer instrumentation framework for diagnosing performance problems in distributed applications, at ACM Symposium on Cloud Computing (SoCC'19)